Wk 4 Supervision Problems 1

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Functions

1. If A is a set with two elements, and B is a set with three elements, how many injective, surjective, and bijective functions are there from A to B? How many of each from B to A?
2. Let $f : X \to Y$ be an injective function. Use f to construct an injective function $F : \mathcal{P}(X) \to \mathcal{P}(Y)$ ($\mathcal{P}$ denotes the power set), justifying why your function is injective.
3. Let $f : X \to Y$ be a surjective function. Write down a set $X' \subset X$ such that $f|_{X'}$ is bijective. *(You may assume the axiom of choice)*

> Axiom of Choice reminder:
> For an arbitrary collection of non-empty sets $\{X_i\}_{i\in I}$, it is possible to select exactly one element from each set as a representative. That is, there exists $\{x_i\}_{i\in I}$ such that $x_i \in X_i$.

Fixed Points of Difference Equations

1. Consider the difference equation

$$x_{n+1} = \lambda x_n - x_n^3,$$

where λ is a non-negative parameter. Find the fixed points of this equation and determine their stability for different values of λ.

2. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function, and consider the difference equation

$$x_{n+1} = f(x_n)$$

Prove (informally) that if this equation has a 2-cycle with the values a and b, then there exists a fixed point $x^* \in (a, b)$.